Research Article | Vol. 16, No. 10 / 1 Oct 2026 / Optical Materials Express 3342
Optical Materials EXPRESS

Energy velocity as an optical design indicator for the ultraviolet nonlinear response of ALD-grown TiO_2 thin films

Diogo Poeta,[1,2] Christian Maibohm,[1] Carlos Calaza,[1] Adelaide Miranda,[1] Martín López-García,[3] and Pieter A. A. De Beule[1,*]

[1]International Iberian Nanotechnology Laboratory, Avenida Mestre José Veiga, s/n, 4715-330, Braga, Portugal
[2]Biomolecular Photonics Research Group, Faculty of Physics, Bielefeld University, Bielefeld, Germany
[3]Instituto de Optica, IO-CSIC, Serrano 121, Madrid, 28006, Spain
*pieter-de.beule@inl.int

Abstract: Slow light media enhance light-matter interactions by increasing the residence time of optical energy, a property with particular relevance for nonlinear nanophotonics. Titanium dioxide (TiO_2) is a promising dielectric material for ultraviolet (UV) applications, yet its strong dispersion and absorption make the commonly used group velocity unreliable as a measure of slow light. Here, we show that the energy velocity (v_e) defined as the ratio of the cycle-averaged Poynting flux to the electromagnetic energy density provides a suitable metric for describing optical transport in TiO_2. We also present the optical and material properties of distinct TiO_2 thin films produced with thermal atomic layer deposition (ALD) with different oxidation agents (H_2O, H_2O_2, O_3). Using O_3 at 200 °C, we obtain fully oxidized, hybrid phase (predominantly amorphous with minor anatase crystallites) TiO_2 films with high refractive index and very small extinction coefficient in the UV-A (315 nm to 400 nm). Ellipsometry combined with analytical estimates and FDTD simulations shows that v_e is strongly reduced, reaching values as low as 0.18 to 0.21 c in the UV-A, confirming that these films operate in a slow-light regime. We experimentally observe that the ALD process chemistry strongly affects the THG response of TiO_2/Si stacks. Because the measurements are performed in reflection on a Si substrate, we interpret the reduced v_e as one optical design indicator among several, rather than as a direct measure of the nonlinear response.

© 2026 Optica Publishing Group under the terms of the Optica Open Access Publishing Agreement

1. Introduction

The capacity to reduce the velocity of light in matter much below the speed of light in vacuum has long been recognized as a powerful tool in nanophotonics [1]. Such light is referred to as slow light and has been exploited in atomic vapors, photonic crystals, and metamaterials to amplify nonlinear response, improve sensing performance, and manipulate information transfer [2–4]. Slow-light significantly enhances light-matter interactions and can sometimes lead to the enhancement of nonlinear optical phenomena [5]. Realizing slow light phenomena in compact, solid state platforms that operate at UV wavelengths remains however a critical challenge.

Several characteristic velocities of light can be defined in dispersive photonic media, including the phase velocity v_p, the group velocity v_g valid for linear, lossless band structures, and the energy velocity v_e describing the speed of energy transport defined as the ratio of the time-averaged Poynting flux to the time-averaged energy density. These velocities are respectively described by

$$v_p = \frac{\omega}{k} \quad ; \quad v_g = \frac{\partial \omega}{\partial k} \quad ; \quad v_e = \frac{\langle S \rangle}{\langle u \rangle}, \tag{1}$$

#611814 https://doi.org/10.1364/OME.611814
Journal © 2026 Received 20 Jul 2026; revised 11 Sep 2026; accepted 14 Sep 2026; published 21 Sep 2026

Optical Materials EXPRESS

whereby ω is the angular frequency and k the angular wavenumber of the light wave under consideration. $\langle \mathbf{S} \rangle$ is the cycle-averaged Poynting vector and $\langle u \rangle$ is the stored electromagnetic energy density [6]. The group velocity becomes ill-defined when absorption is present, exceeding the speed of light in vacuum or even becoming complex [7–9]. In such cases, v_g no longer represents the transport of optical energy and can lead to misleading physical interpretations. TiO_2 offers a high refractive index in the UV-A, making it a suitable dielectric for nanophotonic structures; its finite absorption in this range is precisely why the group velocity becomes unreliable and the energy velocity is the relevant metric.

Intense optical excitation of TiO_2 strongly perturbs its permittivity. Nonlinearities such as the electronic Kerr effect (intensity-dependent refractive index), multi-photon absorption, and free-carrier generation can induce large, ultrafast changes in both the real and imaginary components of the permittivity $\varepsilon(\omega)$. For example, rutile TiO_2 exhibits strong nonlinear absorption near 800 nm, with reported two-photon and three-photon absorption coefficients on the order of 1×10^{-3} cm GW^{-1} and $1\times10\hat{}\{-6\}$ cm^3 GW^{-2} [10]. Such absorption creates free carriers and associated losses, while the Kerr nonlinearity adds a dynamic index shift. These nonlinear processes invalidate a simple linear dispersion relation $k(\omega)$, as the material's band structure dynamically evolves with the optical pump intensity. Consequently, using a small-signal group velocity derived from the unperturbed TiO_2 band structure to represent actual energy flow is generally unreliable. In contrast, the energy velocity v_e inherently tracks how much energy is actually carried by the field. In principle, v_e can be computed from a full electromagnetic model. For instance, finite-difference time-domain (FDTD) simulations can include Kerr and carrier effects by directly evolving Maxwell's equations in a dispersive nonlinear medium [11]. The key point is that v_e includes the effects of absorption and dispersion, by weighting the Poynting flux with the local energy density (including any stored or dissipated energy). This provides an accurate measure of energy propagation speed, even in the presence of material losses and nonlinearities. Crucially, v_e respects causality: it never exceeds the speed of light in vacuum. By contrast, v_g or related group-delay velocities can formally become superluminal in anomalous dispersion or gain-assisted media, but this does not correspond to superluminal information or energy transfer. Frias *et al.* [12] demonstrated this explicitly in a dispersive slab, showing that under strong dispersion conditions ($\varepsilon<1$), the group velocity can exceed c, yet the calculated energy velocity remains below c. Only at a resonance (where the slab is effectively transparent) did v_g and v_e coincide, and only then did the energy velocity approach the vacuum speed of light [12]. This is consistent with general principles of causality and the velocity limit. Therefore, in slow-light nanostructures that exhibit strong nonlinearities and absorption, $v_e = \langle S \rangle / \langle u \rangle$ is the correct metric for quantifying pulse propagation speed.

Titanium dioxide is an abundant material that exists in multiple crystalline phases (rutile and anatase) as well as in an amorphous form. The material is well known for its high refractive index and photocatalytic activity. In photocatalysis, TiO_2 is used to degrade pollutants under UV illumination [13–15], and strategies such as introducing oxygen vacancies or dopants have been employed to extend its activity into the visible range [16,17]. Beyond photocatalysis, TiO_2 has attracted growing interest in nanophotonics due to its compatibility with scalable deposition techniques and its optical properties. Recent reviews cover TiO_2 nanostructures for gas sensing [18] and for environmental photocatalysis [19,20], while further work addresses Bi_2WO_6 nanoparticles on TiO_2 nanotubes for wastewater treatment [21] and defect anchored Pt clusters on TiO_2 for photothermal hydrogen evolution [22]. TiO_2 thin films and metasurfaces have been extensively investigated for both linear and nonlinear optical applications, including waveguides, resonators, and harmonic generation [23,24]. Semmlinger *et al.* demonstrated a TiO_2 metasurface generating third-harmonic light down to the vacuum UV [25]. Amorphous TiO_2 thin films can contain small, randomly oriented anatase crystallites that are optically anisotropic [26], but are nonetheless well described by an isotropic dispersion model [26].

Optical Materials EXPRESS

This approach is consistent with Mueller-matrix ellipsometry results for sol-gel derived TiO_2, whereby no measurable birefringence was observed, even after partial crystallization to anatase [26]. Likewise, studies report that both amorphous and partially crystalline TiO_2 films obtained through Atomic Layer Deposition (ALD) are best modelled with isotropic dispersion functions such as the Tauc-Lorentz model, without the need for anisotropic terms [27,28].

Suppression of far-field scattering and strong field confinement leads to an increased third-harmonic-generation efficiency of a material. This can be achieved through the creation of anapoles or so-called dark modes. Anapole excitation has been demonstrated in various dielectric nanostructures (Si, Ge, etc.) across visible and near-IR wavelengths. Enhanced THG was first reported in Si nanoparticles driven by their magnetic response [29], and non-radiating anapole modes were subsequently demonstrated in dielectric nanoparticles [30]. Anapole excitation has since been used to enhance THG in Ge and Si nanostructures [31,32] and Raman scattering in Si nanodisks [33], for chiral and refractometric sensing [34,35], for mode-locking and ultrafast pulse generation [36], and to achieve high-Q resonances and strong field enhancement in metasurfaces and hybrid nanoantennas [37–40], and even at microwave scales using high-permittivity resonators [41,42]. Anapoles in the UV-A region (350 nm to 380 nm) have been theoretically predicted for high-index amorphous TiO_2 nano-rectangular cuboids (190 nm length, 160 nm width, and 100 nm thickness) deposited atop a fused silica substrate [24]. Experimental demonstration of anapole modes in the UV-A region however remains a significant challenge due to TiO_2 material properties and nanostructuring tolerances. Obtaining TiO_2 films with a thickness of 100 nm, high refractive index, and low extinction coefficient is challenging, as illustrated by the formation of oxygen-deficient "black-TiO_2" when using H_2O as oxidant (see Supplement 1Section 1E). Furthermore, nanosized cuboids require lithography processes creating perfect edges placing very stringent engineering requirements.

Even though nano patterning and the subsequent generation of anapoles can lead to strongly increased THG efficiency, unstructured sputtered nanocrystalline TiO_2 thin films with a thickness of 180 nm have also been demonstrated for the efficient generation of the third-harmonic of 800 nm [43]. It remains an open question how TiO_2 thin films can be employed to efficiently generate VUV light originating from UV-A pump wavelengths through THG. In order to investigate this, we (i) quantify slow-light behaviour in weakly absorbing TiO_2 films through the energy velocity as derived from ellipsometry data combined with FDTD simulations; (ii) show how O_3-grown ALD at 200 °C tunes stoichiometry, crystallinity, and optical constant to realize $v_e \approx 0.18\,c$ in the UV-A; and (iii) compare the third-harmonic response of these films while considering the roles of film thickness, optical constants, and substrate contributions. The THG measurements are performed in reflection on an absorbing Si substrate, so the detected third harmonic is the response of the full TiO_2/Si stack rather than the intrinsic nonlinearity of the film. All optical and nonlinear characterization was performed in air at ambient temperature and pressure; the ALD depositions were carried out at 200 °C and 0.2 Torr as described in Section 2.2.1.

The Supplement 1 provides additional material characterization and numerical support for the conclusions of this work. Specifically, it includes ALD growth calibration and process optimization (Table S1), ellipsometric modelling and depolarization analysis of the TiO_2 films (Table S2; Fig. S1), UV–Vis–NIR reflectance measurements (Fig. S2), AFM morphology and roughness analysis (Fig. S3; Table S3), XPS/REELS/UPS characterization of stoichiometry, defect states, and bandgap trends (Fig. S4–S6; Tables S4–S7), and XRD evidence of partial anatase crystallization (Fig. S7). In addition, the Supplementary Simulations detail the numerical study and optical constants used in the calculations (Table S8), and present energy-velocity maps and UV–Vis–NIR reflectance simulations that complement the main text results (Figs. S9–S11).

Optical Materials EXPRESS

2. Methods

2.1. Energy velocity

2.1.1. Analytical approximation

Assuming a homogeneous, isotropic, linearly absorbing dielectric, we determine v_g using the group index n_g defined by [44]

$$v_g = \frac{c}{n_g} \quad ; \quad n_g = n - \lambda \frac{dn}{d\lambda} \tag{2}$$

We obtain $dn/d\lambda$ from ellipsometry measurements as discussed below. For homogeneous, isotropic, non-magnetic media that are effectively lossless (or only weakly absorbing) and for propagating plane-wave pulses, the energy velocity is approximated by the group velocity, $v_e \approx v_g$. For the Cauchy model $n(\lambda) = A + B/\lambda^2 + C/\lambda^4$, the group index is $n_g(\lambda) = A + 3B/\lambda^2 + 5C/\lambda^4$, with λ in micrometres for the coefficients in Supplement 1Table S2. Table 1 reports the values at the three analysis wavelengths; Fig. 2 retains the n and k curves.

Table 1. Optical constants of the ≈ 98 nm TiO_2 film grown using O_3 (≈ 200 °C), the numerical derivative of the refractive index (computed from spectroscopic ellipsometry), the resulting group index, and the estimated energy velocity. The ratio v_e/c highlights how much the energy velocity is reduced relative to c in vacuum.

λ (nm)	$\approx n$	$\approx k$	$dn/d\lambda$ (nm^{-1})	Group index (n_g)	Energy velocity v_e ($m\,s^{-1}$)	v_e/c
380	2.975	0.019	-0.0056	5.13	$\mathbf{5.85 \times 10^7}$	0.20c
400	2.881	0.014	-0.0044	4.66	$\mathbf{6.44 \times 10^7}$	0.21c
800	2.459	0.00079	-0.0001	2.59	$\mathbf{1.16 \times 10^8}$	0.39c

2.1.2. FDTD simulations

For the rigorous accounting of dispersion and finite absorption in the determination of v_e, the Brillouin energy density valid for dispersive media was determined through finite-difference time-domain (FDTD) simulations [45].

$$\langle u \rangle = \frac{1}{4}\left[\frac{d(\omega\varepsilon'(\omega))}{d\omega}|\mathbf{E}|^2 + \frac{d(\omega\mu'(\omega))}{d\omega}|\mathbf{H}|^2\right] \tag{3}$$

Philbin [8] derived the electromagnetic energy momentum tensor from an action principle, showing that Brillouin's expression is exact in dispersive, lossless media. Luan [46] obtained the same result via power conservation arguments. For absorptive materials, Nunes et al. [47] and Welters et al. [9] extended the framework, demonstrating that the Brillouin energy density corresponds to the portion of energy that can be returned to the field, while the dissipated part contributes to irreversible heating. In practice, the Brillouin form is often adopted as the operational definition of energy density in simulations, with the understanding that it corresponds to the recoverable energy. For nonmagnetic TiO_2 ($\mu' = \mu_0$), the second term reduces to $\mu_0|\mathbf{H}|^2/4$. This expression ensures that the stored energy remains positive and causal even in the presence of strong material dispersion and loss. The FDTD simulations were performed using the open-source package MEEP (MIT Electromagnetic Equation Propagation) [48] running under Ubuntu Linux, as further detailed in section 2 of the supplemental information. MEEP solves Maxwell's equations in the time domain on a Yee grid, with material dispersion incorporated via auxiliary differential equations representing Lorentz-Drude pole extensions of the dielectric function. This approach guarantees causal and Kramers-Kronig consistent modelling of frequency-dependent

Optical Materials EXPRESS

permittivity, allowing accurate reproduction of the measured optical constants. The Lorentz pole parameters were fitted to ellipsometric $n(\lambda)$ and $k(\lambda)$ data and used directly in the simulations.

The complex field distributions at the frequency of interest were recorded using MEEP's Fourier domain monitors, while flux planes provide the net transmitted power. The dispersive permittivity model supplied both $\varepsilon'(\omega)$ and its derivative $d(\omega\varepsilon')/d\omega$, allowing evaluation of the Brillouin energy density voxel-by-voxel. Integration over the simulation domain yields the stored energy, and the ratio of transmitted power to stored energy provides the energy velocity. In the lossless limit, this definition reduces to the usual group velocity, i.e., it coincides with the speed of energy transport for propagating waves. In the presence of absorption, however, the group velocity can become superluminal (v>c/n) or even complex and therefore cannot be interpreted as the velocity of energy transport, whereas $v_e = \langle S\rangle/\langle u\rangle$ remains well defined.

In all simulations, v_e is computed as the ratio of the Fourier-domain Poynting flux to the Brillouin energy density (Eq. (3)), ensuring causality in dispersive, weakly absorptive media. The local energy velocity vector is $\mathbf{v}_e = \langle \mathbf{S}\rangle/\langle u\rangle$, and its scalar magnitude is denoted by $v_e = |\mathbf{v}_e|$. Its direction follows the cycle-averaged Poynting vector; the maps in Fig. 3 show its magnitude together with the energy flow streamlines that indicate its direction. Under normal incidence the linear energy flow is predominantly along $+y$, so the volume averaged scalar values quoted here are representative. The energy velocity analysis is a linear transport calculation and is not vectorially coupled to the third-harmonic polarization.

At wavelengths $\lambda \in \{380\,\text{nm}, 400\,\text{nm}, 800\,\text{nm}\}$, we enforced the target complex index $n(\lambda) + i\,k(\lambda)$ (Table S8) at $\omega_0 = 2\pi c/\lambda$ by setting the real and imaginary parts of the permittivity and the ohmic conductivity so that Meep's linear ohmic medium realizes $\varepsilon_{\text{eff}}(\omega_0) = \varepsilon' + i\,gma/\omega_0$. We have chosen the wavelengths of 380 nm to 400 nm to highlight the impact of absorption onset of TiO_2 moving into the ultraviolet. Additionally, we have added 800 nm as it is a frequently utilized laser wavelength for SHG and THG studies due to the availability of high-power ultrafast laser sources in this wavelength region. For the energy-density calculation, dispersion was taken from the tabulated $n(\lambda)$ and $k(\lambda)$ in Table S8. The volume-averaged v_e was computed over the non-PML (perfectly matched layer) interior of the domain; the reduced $|v_e|$ visible near the lateral edges of the maps corresponds to the absorbing PML and is excluded from the average.

2.2. *TiO_2 thin film*

2.2.1. Fabrication

To investigate the effect of oxidizing agents on the properties of atomic layer deposited (ALD) TiO_2 films, we utilized an ALD system (Beneq TFS200) operated in a cleanroom environment. We chose tetrakis(dimethylamido)titanium (TDMAT) as the Ti precursor for its compatibility with both thermal and plasma enhanced ALD. Three oxidizing agents were compared: deionized water (H_2O), ozone (O_3), and hydrogen peroxide (H_2O_2). All depositions were carried out on 200 mm single-side-polished Czochralski-grown silicon wafers (Siegert Wafer GmbH, prime grade), p-type (boron-doped), (100)-oriented, with a supplier specified resistivity of 1 Ω cm to 100 Ω cm and a thickness of 725(25) µm, at a fixed temperature of 200 °C and a working chamber pressure of 0.2 Torr. Each ALD cycle consisted of a TDMAT pulse, followed by a nitrogen purge, then an oxidant pulse, and a final purge step. For each oxidant (H_2O, O_3, H_2O_2), we performed 1250 ALD cycles to grow TiO_2 films. Additionally, using O_3 we deposited a thicker film with 2135 cycles targeting a 100 nm thin film. The measured thickness of approximately 98 nm obtained after 2135 cycles corresponds to a growth per cycle of approximately 0.46 Å, compared with approximately 0.47 Å for the 59 nm film grown with 1250 cycles under the same conditions. These values are close to the approximately 0.46 Å per cycle reported by Jin *et al.* for TDMAT/O_3 thermal ALD within the 100 °C to 250 °C growth window [49].

The Al_2O_3 depositions using trimethylaluminium (TMA) and O_3 were not used to establish the TiO_2 growth window. They served only to verify that ozone reached the chamber at the

Optical Materials EXPRESS

intended flow, since our chamber volume is larger than in the prior study on which we based our process [50]. Al_2O_3 from TMA and O_3 is a reproducible process and therefore a convenient reporter for oxidant delivery. Reflectometry data (Hyperion NanoCalc) showed that varying the O_3 flow rate from 12 L min^{-1} to 40 L min^{-1} on Si wafers nearly doubled the Al_2O_3 thickness for a given number of cycles, demonstrating a strong oxidant flow dependence as documented in the Supplemental Information. The O_3 flow identified in this way was then applied to the TiO_2 depositions. The TiO_2 process conditions themselves, temperature, pressure, pulse and purge times, were set independently.

2.2.2. Physical, chemical, and optical characterization

Material properties of the TiO_2 films are characterized through spectroscopic ellipsometry (SE), ultraviolet–visible–near–infrared (UV–Vis–NIR) spectroscopy, atomic force microscopy (AFM), X-ray photoelectron spectroscopy (XPS) and X-ray diffraction (XRD). SE determines TiO_2 film thickness and refractive index by fitting $\Psi(\lambda)$ and $\Delta(\lambda)$ spectra between 370 nm to 1000 nm using a Cauchy dispersion model appropriate for transparent dielectrics. Optical reflectance was evaluated through UV-Vis-NIR using (Lambda 950, PerkinElmer). Specular reflectance was measured from 300 nm to 800 nm using a variable-angle relative reflectance accessory at 45° incidence for the comparison between oxidants; the thick O_3-grown film was additionally measured at 30°, 45° and 60°. Prior to each measurement, the system was calibrated using a mirror at 45° incidence to establish a baseline, ensuring accurate comparison across all wavelengths. Because the Si substrate is opaque in the UV–visible range, conventional transmission absorbance of the film cannot be obtained from these samples. An estimate $A \approx 1 - R$ would describe absorptance of the full stack and requires total reflected power, or negligible uncollected scattering when using specular reflectance. It would not isolate absorption in TiO_2; a direct film comparison would require samples on a transparent substrate. The investigated samples can be described as an asymmetric Fabry–Pérot-type thin-film stack consisting of air/TiO_2/Si [51]. In the UV-A spectral range considered here, the Si substrate is strongly absorbing and can therefore be treated as optically semi-infinite, such that the relevant optical response is determined by the air/TiO_2/Si stack. This stack determines the interference conditions and the field distribution inside the TiO_2 layer, and is therefore also relevant to the THG response discussed below. AFM topography scans were performed in intermittent contact (tapping) mode in air (JPK Nanowizard 3 AFM, Bruker Nano GmbH, Berlin, Germany). Silicon nitride cantilevers with pyrex-nitride tips (PPP-NCHAu, Nanosensors) were used; they had a nominal resonance frequency of 330 kHz in air and a spring constant of 29 N m^{-1}. The AFM instrument is mounted on a noise vibration platform (i4, Accurion GmbH, Göttingen, Germany) and the entire set-up is installed on an ultra-low noise floor constructed to comply with the NIST-A vibration specifications [52]. The system has a demonstrated noise floor of ≈ 75 pm [53]. All investigations were carried out with out-of-the-box cantilevers. Images with a scan size of (2 μm × 2 μm) were acquired in three different areas for each sample. Image analysis and determination of the roughness parameters were performed using the JPK Data Processing software (version 6.1.116). Elemental composition and oxidation states were assessed via XPS using a Thermo Fisher ESCALAB 250Xi instrument with a Mg Kα source. High-resolution spectra were recorded with a 1.0 eV step size. Deconvoluting these spectra enabled quantification of Ti3+ content and assessment of oxygen stoichiometry. Analyzed samples included TiO_2 films grown with H_2O, H_2O_2 and O_3 oxidants. Films grown using H_2O as the oxidant resulted in visibly darker films, referred to as *black* TiO_2 [50]. XRD measurements were carried out using a PANalytical X'Pert PRO diffractometer in Bragg–Brentano geometry, with a Cu anode X-ray source (Kα_1 = 1.540598 Å, Kα_2 = 1.544426 Å, intensity ratio = 0.5). Scans were performed in continuous mode over a 2θ range of 10° to 80° with a step size of 0.0098°, using a fixed divergence slit (0.76 mm). The generator was operated at 45 kV and

Optical Materials EXPRESS

40 mA. A fine calibration offset of −0.0193° 2θ was applied. No monochromator was used during measurement.

2.2.3. THG measurements

THG measurements were performed in reflection using the optical arrangement shown in Fig. 1. Femtosecond pulses from a Carbide CB5 (Light Conversion, 1030 nm, 60 KHz, 6 W, 100 uJ) laser pumping an Orpheus optical parametric amplifier (Light Conversion) provided the used tunable infrared radiation in the 1110 nm to 1350 nm range. The beam was steered by protected silver mirrors (*E*1, *E*2, Thorlabs, PF10-03-P01) onto a reflective neutral density filter (*M*, Thorlabs, ND03B), which attenuated the pump and directed it through an uncoated plano convex N-BK7 lens (*L*, Thorlabs, LA1252) onto the sample. The sample was mounted on an XYZ translation stage to position the focus on the TiO_2 and Si samples. The THG signal was collected in EPI configuration by the same lens (*L*). In the detection path the residual pump was suppressed with a single-band bandpass filter (*BP*, Semrock, FF01-439/154-25, 439+/-77 nm) tuned to the third-harmonic wavelength, and the filtered signal was coupled through an Olympus 4× objective (*O*, Olympus, RMS4X, WD 18.5 mm, NA 0.1) into a fiber coupled spectrometer (AvaSpec ULS2048×64 EVO). The beam transmitted through the reflective neutral-density filter was blocked in a beam dump. All experiments were conducted in air at a temperature of 20 °C. Collection was at normal incidence to the sample surface. The transmitted third harmonic was not measured, since the Si substrate is opaque in this spectral range. We observed no additional spectral features within the detection band; emission at other wavelengths was not measured, since the detection path is filtered around the third harmonic. The THG spectra are compared as relative intensities. The measurements were not calibrated against an absolute radiometric standard, and the raw acquisitions are no longer available for a retrospective repeatability analysis. Consequently, no THG error bars or complete uncertainty budget can be provided, and the

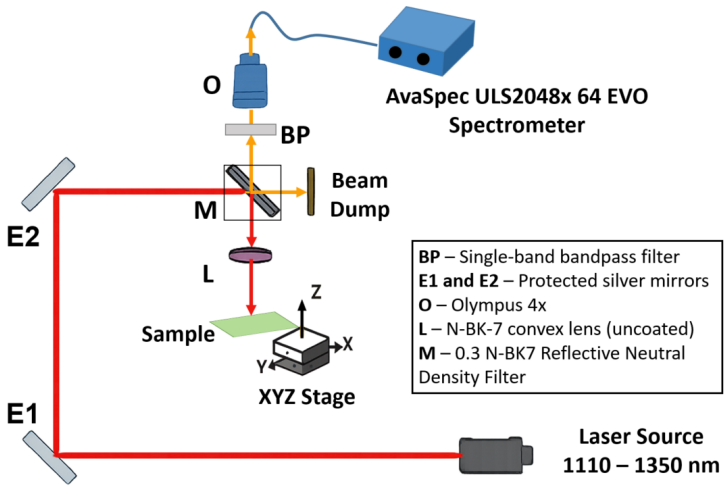


Fig. 1. Experimental setup for third-harmonic measurements. A tunable infrared source (1110 nm to 1350 nm) is directed by protected-silver mirrors (*E*1, *E*2) onto a reflective neutral density filter (*M*),then focused by an uncoated N-BK7 lens (*L*) onto the sample on an XYZ stage. The reflected third-harmonic is collected by (*L*), spectrally filtered by a bandpass filter (*BP*), coupled through an Olympus 4× objective (*O*), and recorded by an AvaSpec ULS2048×64 EVO spectrometer. The pump is normally incident on the sample, and the third harmonic is collected backwards through the same lens. A photograph of the setup is shown in Supplement 1Fig. S8.

Optical Materials EXPRESS

statistical significance of differences between samples cannot be established from these spectra. The roughness standard deviations in Supplement 1Table S3 and the fit uncertainties in Table S2 describe those measurements only; they do not quantify the THG uncertainty.

An experimental irradiance dependence series was not acquired for every sample. Normalization by P^3_{pump} therefore assumes the expected cubic THG scaling rather than establishing it experimentally for each film. The numerical source amplitude sweep described in Supplement 1Section 2 is a model check and does not validate the experimental power dependence. The measurements do not provide absolute conversion efficiencies.

3. Results

Spectroscopic ellipsometry was performed on the four different TiO_2 thin film ALD samples detailed in Section 2.2.1. Detailed ellipsometric modelling, fit parameters, and depolarization analysis are provided in the supplemental information Table S2 and Fig. S1. The ellipsometry derived thickness confirmed the ALD growth trends: ≈ 41.99 nm for H_2O-grown film, ≈ 58.57 nm for O_3-grown film, and ≈ 52.87 nm for H_2O_2-grown film (all at 1250 cycles), and ≈ 98 nm for the thicker O_3 film (2135 cycles). A stronger oxidant produces a thicker film for the same number of ALD cycles, consistent with a higher reactivity promoting faster growth. Based on the ellipsometry data, the energy velocity analysis is presented in Table 1 for the 98 nm TiO_2 slab, corresponding to the O_3-grown film in 2135 ALD cycles. This film thickness corresponds to the target thickness required for UV-A anapole excitation [24]. Results indicate that the energy velocity is reduced to ≈ 0.20 c at 380 nm and ≈ 0.21 c at 400 nm, while assuming a value of ≈ 0.39 c at 800 nm. The UV-A values are evaluated at the detected third-harmonic wavelengths; the 800 nm point is a separate reference wavelength. The nonlinear polarization is driven by the field at the fundamental (1140 nm to 1200 nm), where extrapolation of the Cauchy fit beyond the measured 370–1000 nm range gives $v_e/c \approx 0.41$. This is an analytical estimate rather than an independent measurement at the pump wavelengths. The reduced v_e near 380 nm to 400 nm instead affects propagation, absorption and extraction of the generated harmonic [54]. Figure 2 displays the optical constant dispersion data extracted from spectroscopic ellipsometry of the O_3-grown TiO_2 film, reporting a high refractive index of approximately $n \approx 3.0$ at 380 nm and a very small extinction coefficient k. This indicates that the film is both optically dense and highly transparent in the UV-A, which is advantageous for enhancing nonlinear interactions without significant absorption losses.

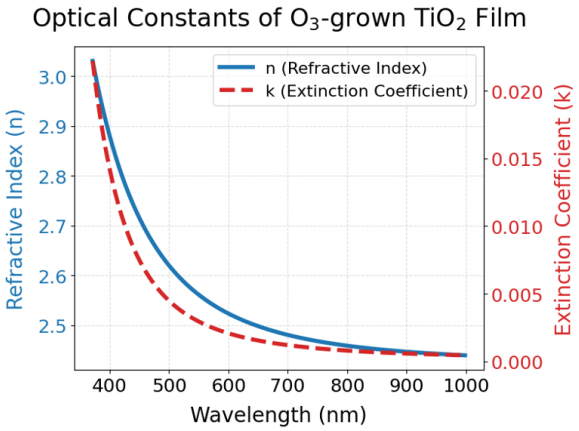


Fig. 2. Ellipsometry data fit for the refractive index n (blue) and extinction coefficient k (red) for the O_3-grown TiO_2 film.

Optical Materials EXPRESS

FDTD simulations have been performed for 100 nm thick slabs of TiO_2, Si, and SiO_2 embedded in air, illuminated at normal incidence with transverse-electric (TE_z) polarization, as depicted in Fig. 3 **a)**. 2D domain spanning 4 μm in-plane with 1 μm of air above and below the film and 1 μm perfectly matched layers (PMLs) on all boundaries have been considered. The spatial resolution was 20 pixels per 100 nm, corresponding to a grid spacing of 5 nm. A Gaussian-pulse plane wave centered at the analysis wavelength λ_0 is launched from the lower air region, and the energy velocity is evaluated at the single analysis frequency $f_0 = c/\lambda_0$ using the Fourier-domain monitors. Each wavelength $\lambda_0 \in \{380\,\text{nm}, 400\,\text{nm}, 800\,\text{nm}\}$ is simulated independently, so the source is narrowband at f_0 rather than broadband.

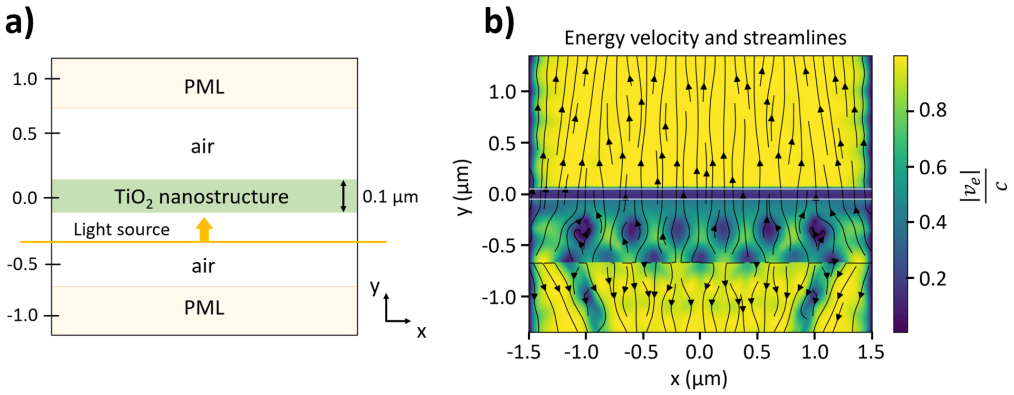


Fig. 3. (a) Simulation layout for the 2D FDTD model (Meep): a 100 nm TiO_2 film is centered at $y = 0$, surrounded by air and terminated by PMLs. A TE_z line source in the lower air region launches a normally-incident plane wave in the $+y$ direction. (b) Energy velocity map and streamlines at $\lambda_0 = 380$ nm. Colors show $|v_e|/c$ computed as $v_e = \langle S \rangle / \langle u \rangle$. Streamlines indicate energy flowing upward from the source through the slab into the upper half space. The reduced $|v_e|/c$ near the left and right edges of the map corresponds to the 1 µm absorbing PML at the lateral boundaries and is not physically significant.

Figure 3 **b)** indicates v_e/c computed via $v_e = \langle S \rangle / \langle u \rangle$ using the cycle-averaged Poynting vector $S = \frac{1}{2}\text{Re}\{E \times H^*\}$ and the Brillouin energy density $u = \frac{1}{4}\left[\partial_\omega(\omega\varepsilon')|E|^2 + \partial_\omega(\omega\mu')|H|^2\right]$. In air, one observes $v_e \approx c$, while dispersion and absorption reduce v_e well below c inside the TiO_2 slab, consistent with the definition and with Meep's frequency-domain flux formalism (Poynting flux of the Fourier-transformed fields) and Poynting's theorem. Dispersion and the small but finite absorption reduce the local energy velocity well below c, with interior regions at $\lambda_0 = 380$ nm showing $v_e/cmeq0.2$ (see also Fig. S9 for the 800 nm TiO_2 map and Fig. S10 for the air/SiO_2 maps). Volume averaged values extracted from the FDTD simulations yield $v_e/c = 0.185$ at 380 nm, 0.201 at 400 nm, and 0.328 at 800 nm, in good agreement with the analytical estimates $v_e/c \approx 0.20$, 0.21 and 0.39 reported in Table 1. v_e is systematically lower than the corresponding group velocity ($v_g/c = 0.195$, 0.216, 0.386), so $v_e/v_g \approx 0.95$, 0.93 and 0.85, respectively, giving a calculated difference of 5–15 %. We note that a difference of this magnitude is comparable to the offset introduced by the volume averaging window, which is visible in the SiO_2 control (Supplement 1Section 2A). The calculated v_e values are lower than the corresponding v_g estimates, but the comparable averaging offset prevents us from treating the full difference as a resolved physical effect. The present 5–15 % difference therefore cannot be used to predict a corresponding change in THG or prescribe a film thickness correction.

Figure 4 summarizes the normalized phase, group, and energy velocities, v_p/c, v_g/c, and v_e/c, for TiO_2, Si, and SiO_2 at 380 nm, 400 nm and 800 nm (same 100 nm slab, TE_z, normal incidence). Two trends stand out: (i) the computed v_e values for TiO_2 and Si lie below the corresponding v_g

Optical Materials EXPRESS

estimates, subject to the averaging limitation discussed above; (ii) for SiO_2, with very small k and comparatively weak dispersion in this band, v_e and v_g track more closely, so v_g is often an adequate descriptor for transport in low-loss silica films. Supplement 1Figs. S9 and S10 show v_e/c maps and streamlines across UV-NIR wavelengths for a 100 nm slab. As dispersion and absorption increases (towards the UV or in higher-index media), the spatial distribution of v_e becomes highly non-uniform and v_e falls below c, subject to the averaging limitation discussed above. Control maps in air and in a 100 nm SiO_2 slab (Fig. S10) show: air gives $v_e/c \approx 1$ while low-loss SiO_2 shows $v_e/c \approx 0.585$ and $v_e \approx v_g$, emphasizing that light velocity within SiO_2 is equivalent to v_g [55], while TiO_2 or Si require v_e to accurately describe light velocity within.

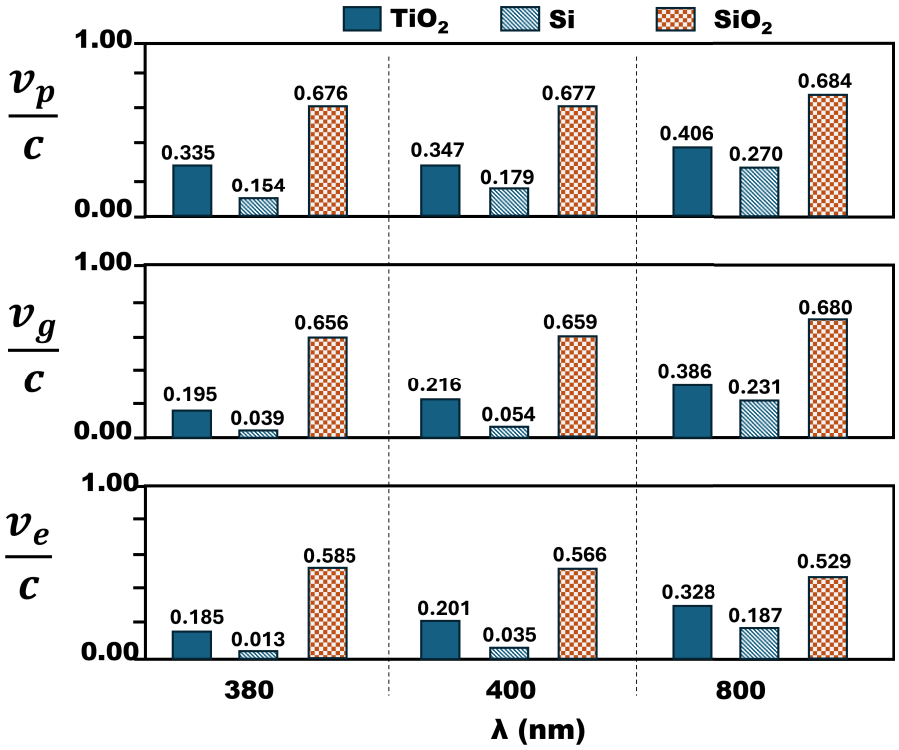


Fig. 4. Normalized velocities for 100 nm slab under TE_z excitation at normal incidence, evaluated at λ = 380; nm400; nm 800 nm. The three panels show (top) phase velocity v_p/c, (middle) group velocity v_g/c, and (bottom) energy velocity v_e/c. The horizontal axis is wavelength λ (nm) and the vertical axis is the corresponding normalized velocity. Bar colour and hatching indicate the material: solid blue = TiO_2, hatched blue = Si, and orange cross = SiO_2. Numeric labels give the values of v/c.

Figure 5 shows third-harmonic spectra from bare Si and TiO_2/Si samples grown with different oxidants in the UV-A wavelength region. A narrow peak is observed at the expected $\lambda_{3\omega}$, confirming third harmonic frequency conversion. The peak amplitude depends strongly on film chemistry and thin film thickness. Because bare Si produces a clear third harmonic signal, these spectra represent the response of the complete TiO_2/Si stack rather than the intrinsic nonlinearity of TiO_2. The H_2O_2-grown TiO_2 film and the thicker O_3-grown TiO_2 film (2135 cycles, ≈ 98 nm) exhibit the strongest THG signals at $\lambda_{3\omega}$ = 380 nm, whereas the H_2O-grown film yields the weakest response. The THG response at $\lambda_{3\omega}$ = 400 nm differs significantly, whereby the H_2O_2-grown film remains highest followed by the thinner O_3-grown film (1250 cycles). Here, the thick O_3 film and bare Si provided intermediate signal strength. To enable a

Optical Materials EXPRESS

direct comparison between the two excitation settings, the third-harmonic spectra in Fig. 5 were normalized by the cube of the measured pump power at each excitation wavelength, $P^3_{\text{pump}}(\lambda)$, following the cubic power dependence of THG. Within each panel all samples were measured at the same wavelength and pump power, so this normalization is a common factor and does not affect the ranking within a panel; it matters only when comparing between panels. The fundamental wavelengths for panels **a)** and **b)** were 1140 nm and 1200 nm, respectively. This normalization therefore introduces only a minor correction and does not affect the relative ordering between samples.

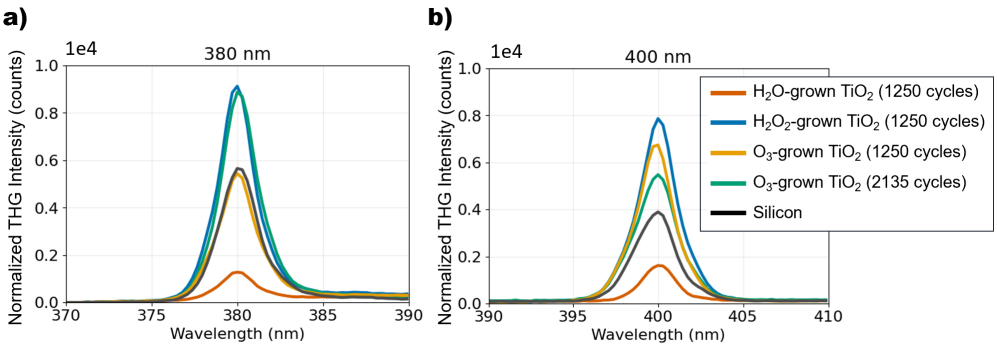


Fig. 5. Third-harmonic spectra recorded from bare silicon and TiO_2 coated samples **a)** $\lambda_\omega = 1140$ nm and $\lambda_{3\omega} = 380$ nm . **b)** $\lambda_\omega = 1200$ nm and $\lambda_{3\omega} = 400$ nm.

Ellipsometry modelling of the H_2O_2-grown TiO_2 thin film is complicated by a non-negligible absorption that is difficult to parametrize reliably, unlike the O_3 film, for which the Cauchy + Urbach model fits cleanly. Hence, one cannot analytically extract the energy velocity for this film with the same confidence as for the O_3-grown TiO_2 thin film. We attribute its strong THG response qualitatively to the combined effect of its intermediate refractive index, its film thickness (≈ 53 nm), and its defect chemistry as detailed below, while noting that resolving this quantitatively requires a more complete optical model. Comparably reliable optical constants could not be obtained for all films, so we do not report v_e for every film in Table 2 or plot THG peak intensity against v_e. Such a comparison requires reliable optical models for the absorbing films and a sample series that separates thickness, composition and substrate interference. In reflection geometry, the measured response originates from an asymmetric Fabry–Pérot-type air/TiO_2/Si stack, in which the TiO_2 layer and the absorbing Si substrate determine the interference and field-enhancement conditions. Since measurements are performed on TiO_2/Si in reflection geometry, the detected third-harmonic should be interpreted as the response of the full TiO_2/Si stack: the Si substrate can contribute directly, and the TiO_2 layer modifies both the pump field distribution and the absorption conditions for the generated harmonic.

Table 2 summarizes material properties most relevant for the interpretation of the third-harmonic response derived from SE, AFM, XRD and XPS. The H_2O-grown film exhibits the clearest defect signature, while the O_3-grown films are closest to stoichiometric TiO_2. The H_2O_2-grown film shows intermediate material properties but yields the strongest measured third-harmonic signal, indicating that the observed response is not determined only by stoichiometry, but also by the combined effects of optical dispersion, defect chemistry, and field distribution in the TiO_2/Si structure. Interestingly, we observe the anatase (101) peak in O_3-grown TiO_2 films deposited at only 200°C, even without any post-deposition annealing. We attributed this partial crystallization to the high oxidative reactivity of ozone. The O_3 process likely facilitates more complete ligand removal and Ti–O bond formation during deposition, providing enough local energy for nuclei of anatase to form. These crystallites seem to nucleate at the surface

Optical Materials EXPRESS

Table 2. Material characterization results for the TiO_2 films used in the third-harmonic-generation study. Film thickness values are approximate.

Oxidant	ALD cycles	Film thickness (nm)	R_q (nm)	O/Ti	Ti^{3+} presence	REELS bandgap (eV)	THG relevance
H_2O	1250	42	0.79	1.95	Yes	3.558	weakest response
H_2O_2	1250	53	0.45	1.60	moderate	3.531	strongest response
O_3	1250	59	0.42	1.99	No	3.571	strong response
O_3	2135	98	1.88	2.00	No	3.452	low-energy velocity regime

(where atomic mobility and reaction likelihood are highest) while the bulk of the film remains amorphous, resulting in a hybrid phase. A similar phenomenon was reported by Jin *et al.*, [49], who observed anatase formation in ALD TiO_2 at low temperatures using O_3. The supplemental material provides an in-depth study of the different oxidants.

The materials data show that the ALD oxidant controls stoichiometry, roughness, and (partial) crystallinity, which jointly determine the optical constants and therefore the energy-transport properties of the films. O_3-grown TiO_2 is closest to stoichiometric TiO_2 (no clear Ti^{3+} signature) and exhibits a high refractive index with only small extinction in the UV-A, while H_2O-grown films show Ti^{3+} related defect signatures. The thicker O_3 film additionally shows increased roughness and optical features near the band edge (Supplement 1Fig. S2), consistent with a change in the internal field distribution and light trapping in the UV-A. Defect chemistry therefore enters the present analysis through the measured optical constants: oxygen-deficient, Ti^{3+}-containing films show increased sub-bandgap absorption, which raises k in the UV-A and lowers the recoverable energy density entering v_e. No controlled defect density series was performed, so the present data do not isolate a quantitative contribution of defects to v_e or THG. Laser induced structural modification of amorphous TiO_2 has likewise been shown to alter its nonlinear optical response [56].

The THG measurements (Fig. 5) show that the detected third harmonic depends sensitively on the film chemistry and thickness. The H_2O_2-grown film provides the strongest signal at $\lambda_{3\omega} = 400\,\text{nm}$, while the thicker O_3-grown film becomes competitive at $\lambda_{3\omega} = 380\,\text{nm}$. This wavelength-dependent ranking indicates that the measured signal is shaped by multiple factors beyond stoichiometry alone, including (i) the linear optical response of the TiO_2/Si stack (interference and field enhancement at both ω and 3ω), (ii) absorption and scattering losses, (iii) the nonlinear contribution of the Si substrate, and (iv) the differing film thicknesses, which alter both the interaction length and the interference condition at ω and 3ω. In particular, the H_2O_2-grown film is thinner than the O_3-grown film measured at 1250 cycles, so the observed ranking cannot be attributed to optical transport alone. The non-negligible third harmonic from bare Si in Fig. 5 reinforces that the reflection geometry probes the full stack rather than the film in isolation. In this configuration, the TiO_2 layer can enhance or suppress the detected third harmonic by modifying the pump field distribution in Si and by changing the extraction conditions for the generated harmonic. Therefore, while reduced v_e near the UV band edge is a useful design indicator for enhanced light-matter interaction in TiO_2, a direct, quantitative link between v_e and an intrinsic film $\chi^{(3)}$ cannot be established from reflection on Si spectra alone. Hence, future work will focus on suppressing the substrate contribution and quantify the film nonlinearity on an optically transparent substrate such as fused silica, using transmission and reflection geometries with wavelength dependent normalization by P^3_{pump} and calibrated

Optical Materials EXPRESS

detection. Combined with the energy-velocity framework, such measurements will enable a thickness and dispersion optimization strategy for UV-range nonlinear conversion in ALD-grown TiO_2, both in simple slabs and patterned resonators.

4. Conclusion

We established energy velocity $v_e = \langle S \rangle / \langle u \rangle$ as a suitable metric to describe optical transport in the UV-A wavelength range for TiO_2, where strong dispersion and finite absorption render the well-known group velocity unreliable for estimation of material's optical nonlinearity. Energy velocity has previously been used to describe optical energy transport in dispersive media [7], and thermal ALD of TiO_2 using different oxidants is well established [49,50]. The contribution here is their combination: quantifying v_e for ALD-grown TiO_2 in the UV-A from measured optical constants, in the band where the absorption onset makes v_g unreliable, and connecting it to a nonlinear measurement on the same films. O_3-grown thermal ALD at 200 °C yields high-index TiO_2 thin films with small extinction in the UV-A and enables an energy-transport slowdown to $v_e/c \approx 0.18$–0.21 around $\lambda = 380\,\text{nm}$–$400\,\text{nm}$, confirmed by ellipsometry and FDTD. Experimentally, reflected THG spectra from TiO_2/Si stacks exhibit intensities that depend strongly on oxidant chemistry and thickness. The H_2O_2-grown film produces the strongest signal at 400 nm, while the thicker O_3-grown film becomes stronger at 380 nm. The third harmonic signal from bare Si indicates that reflection measurements report a coupled film-substrate response, emphasizing the need for substrate suppressed geometries to extract intrinsic film nonlinear coefficients. In summary these results indicate that ALD process chemistry can be utilized to engineer both the UV-A energy transport regime and the observed nonlinear response of TiO_2 nanostructures. Because v_e follows directly from the ellipsometric $n(\lambda)$ and $k(\lambda)$, the analytical estimate carries essentially no computational cost, and the FDTD maps are lightweight two-dimensional single-frequency calculations rather than demanding simulations. The approach is therefore practical for screening film thickness, oxidant chemistry, and operating wavelength against a target slow-light band prior to fabrication. Because v_e is computed directly from the simulated fields, the same FDTD analysis can be applied to patterned geometries, including the UV-A anapole supporting TiO_2 cuboids for which this film thickness was targeted [24], although interpreting a volume averaged v_e requires more care in strongly resonant structures. Its advantage over the group velocity is that it remains causal and bounded by c in the absorbing, strongly dispersive UV-A band, where v_g can exceed c or become complex. The main limitation is that v_e describes linear energy transport, so it identifies favorable conditions for light-matter interaction but does not by itself yield an intrinsic $\chi^{(3)}$, and its accuracy near the band edge depends on a well constrained extinction coefficient. Coupling the energy velocity analysis with fully nonlinear dispersive FDTD that includes Kerr and free-carrier terms, together with substrate suppressed measurements on a transparent substrate, would extend the present design indicator into a quantitative thickness and dispersion optimization for UV harmonic generation. Beyond the films studied here, these results are relevant for nonlinear UV devices, including compact solid-state harmonic sources for photoionization and spectroscopy [57], and for optical modulation and beam shaping using the Kerr response of nanostructured TiO_2 [58].

Funding. European Commission (101057510).

Acknowledgments. The authors thank Michael Scalora from Weapons Sciences Directorate Research, Development and Engineering Center for insightful discussions on the theoretical formulation of energy velocity in dispersive media and for critical comments on this research. We thank the Micro and Nanofabrication Facility at INL for their support on the development of the TiO_2 films, in particular Helder Fonseca and Andrea Gouvêa, and the Electron Microscopy and X-Rays Facility at INL, in particular, Oleksandr Bondarchuk for the XPS measurement and Elvira Paz for the XRD measurement. We also acknowledge funding support from the EU project LEARN (Development of novel assessments for indoor air quality monitoring and impact on children's health), Grant No. 101057510 under Horizon Europe (HORIZON-HLTH-2021-ENVHLTH-02).

Disclosures. The authors declare no conflicts of interest.

Optical Materials EXPRESS

Data Availability. Data underlying the results presented in this paper are available from the corresponding author upon reasonable request.

Supplemental document. See Supplement 1 for supporting content.

References

1. T. F. Krauss, "Why do we need slow light?" Nat. Photonics **2**(8), 448–450 (2008).
2. D. Budker and M. Romalis, "Optical magnetometry," Nat. Phys. **3**, 227–234 (2006).
3. T. Baba, "Slow light in photonic crystals," Nat. Photonics **2**(8), 465–473 (2008).
4. S. Zhang, D. Genov, Y. Wang, *et al.*, "Plasmon-induced transparency in metamaterials," Phys. Rev. Lett. **101**(4), 047401 (2008).
5. R. W. Boyd, "Material slow light and structural slow light: similarities and differences for nonlinear optics [Invited]," J. Opt. Soc. Am. B **28**(12), A38 (2011).
6. M. Scalora, M. A. Vincenti, D. de Ceglia, *et al.*, "Extreme electrodynamics in time-varying media," Phys. Rev. A **112**(1), 013502 (2025).
7. G. C. Sherman and K. E. Oughstun, "Energy-velocity description of pulse propagation in absorbing, dispersive dielectrics," J. Opt. Soc. Am. B **12**(2), 229–247 (1995).
8. T. G. Philbin, "Electromagnetic energy momentum in dispersive media," Phys. Rev. A **83**(1), 013823 (2011).
9. A. Welters, Y. Avniel, and S. G. Johnson, "Speed-of-light limitations in passive linear media," Phys. Rev. A **90**(2), 023847 (2014).
10. C. C. Evans, J. D. B. Bradley, E. A. M.-P. no, *et al.*, "Mixed two- and three-photon absorption in bulk rutile (tio2) around 800 nm," Opt. Express **20**(3), 3118–3128 (2012).
11. R. Joseph and A. Taflove, "Fdtd maxwell's equations models for nonlinear electrodynamics and optics," IEEE Trans. Antennas Propag. **45**(3), 364–374 (1997).
12. W. Frias, A. Smolyakov, and A. Hirose, "Non-local energy transport in tunneling and plasmonic structures," Opt. Express **19**(16), 15281–15296 (2011).
13. M. Rosales, T. Zoltan, C. Yadarola, *et al.*, "The influence of the morphology of 1d tio2 nanostructures on photogeneration of reactive oxygen species and enhanced photocatalytic activity," J. Mol. Liq. **281**, 59–69 (2019).
14. F. Zhou, C. Yan, Q. Sun, *et al.*, "Tio2/sepiolite nanocomposites doped with rare earth ions: Preparation, characterization and visible light photocatalytic activity," Microporous Mesoporous Mater. **274**, 25–32 (2019).
15. W. H. Abdelraheem, M. K. Patil, M. N. Nadagouda, *et al.*, "Hydrothermal synthesis of photoactive nitrogen- and boron- codoped tio2 nanoparticles for the treatment of bisphenol a in wastewater: Synthesis, photocatalytic activity, degradation byproducts and reaction pathways," Appl. Catal. B **241**, 598–611 (2019).
16. J. Li, M. Zhang, Z. Guan, *et al.*, "Synergistic effect of surface and bulk single-electron-trapped oxygen vacancy of tio2 in the photocatalytic reduction of co2," Appl. Catal. B **206**, 300–307 (2017).
17. T. R. Gordon, M. Cargnello, T. Paik, *et al.*, "Nonaqueous synthesis of tio2 nanocrystals using tif4 to engineer morphology, oxygen vacancy concentration, and photocatalytic activity," J. Am. Chem. Soc. **134**(15), 6751–6761 (2012).
18. G. W. C. Kumarage, H. Hakkoum, and E. Comini, "Recent advancements in tio2 nanostructures: Sustainable synthesis and gas sensing," Nanomaterials **13**(8), 1424 (2023).
19. M.-A. Gatou, A. Syrrakou, N. Lagopati, *et al.*, "Photocatalytic tio2-based nanostructures as a promising material for diverse environmental applications: A review," Reactions **5**(1), 135–194 (2024).
20. A. Jayaseelan, N. Senthilnathan, G. Rangarajan, *et al.*, "Synthesis and application of titanium dioxide photocatalysis for energy, decontamination and viral disinfection: a review," Environ. Chem. Lett. **21**(1), 339–362 (2022).
21. Q. Wang, H. Li, X. Yu, *et al.*, "Morphology regulated bi2wo6 nanoparticles on tio2 nanotubes by solvothermal sb3+ doping as effective photocatalysts for wastewater treatment," Electrochim. Acta **330**, 135167 (2020).
22. D. Jiang, H. Yuan, Z. Liu, *et al.*, "Defect-anchored single-atom-layer pt clusters on tio2-x/ti for efficient hydrogen evolution via photothermal reforming plastics," Appl. Catal. B **339**, 123081 (2023).
23. C. C. Evans, C. Liu, and J. Suntivich, "Low-loss titanium dioxide waveguides and resonators using a dielectric lift-off fabrication process," Opt. Express **23**(9), 11160–11169 (2015).
24. D. Poeta, R. Vilarinho, M. López-García, *et al.*, "Optical anapoles excited by uv-a illumination," Opt. Lett. **49**(24), 7198–7201 (2024).
25. M. Semmlinger, M. Zhang, M. L. Tseng, *et al.*, "Generating third harmonic vacuum ultraviolet light with a tio2 metasurface," Nano Lett. **19**(12), 8972–8978 (2019).
26. J. Niziol, M. Zieba, M. Sniechowski, *et al.*, "Demonstration of the optical isotropy of tio2 thin films prepared by the sol-gel method," Materials **17**(14), 3391 (2024).
27. O. M. Ylivaara, A. Langner, X. Liu, *et al.*, "Mechanical and optical properties of as-grown and thermally annealed titanium dioxide from titanium tetrachloride and water by atomic layer deposition," Thin Solid Films **732**, 138758 (2021).
28. D. Saha, R. Ajimsha, K. Rajiv, *et al.*, "Spectroscopic ellipsometry characterization of amorphous and crystalline tio2 thin films grown by atomic layer deposition at different temperatures," Appl. Surf. Sci. **315**, 116–123 (2014).
29. M. R. Shcherbakov, D. N. Neshev, B. Hopkins, *et al.*, "Enhanced third-harmonic generation in silicon nanoparticles driven by magnetic response," Nano Lett. **14**(11), 6488–6492 (2014).

Optical Materials EXPRESS

30. A. Miroshnichenko, A. Evlyukhin, Y. F. Yu, *et al.*, "Nonradiating anapole modes in dielectric nanoparticles," Nat. Commun. **6**(1), 8069 (2015).
31. G. Grinblat, Y. Li, M. Nielsen, *et al.*, "Enhanced third harmonic generation in single germanium nanodisks excited at the anapole mode," Nano Lett. **16**(7), 4635–4640 (2016).
32. L. Wang, S. Kruk, L. Xu, *et al.*, "Shaping the third-harmonic radiation from silicon nanodimers," Nanoscale **9**(6), 2201 (2017).
33. D. Baranov, R. Verre, and P. Karpinski, "Anapole-enhanced intrinsic raman scattering from silicon nanodisks," ACS Photonics **5**(7), 2730–2736 (2018).
34. G. Serrera, J. González-Colsa, and P. Albella, "Amplified linear and nonlinear chiral sensing assisted by anapole modes in hybrid metasurfaces," Appl. Phys. Lett. **124**(25), 251701 (2024).
35. J. F. Algorri, D. C. Zografopoulos, A. Ferraro, *et al.*, "Anapole modes in hollow nanocuboid dielectric metasurfaces for refractometric sensing," Nanomaterials **9**(1), 30 (2019).
36. J. S. Totero Gongora, A. Miroshnichenko, Y. Kivshar, *et al.*, "Anapole nanolasers for mode-locking and ultrafast pulse generation," Nat. Commun. **8**(1), 15535 (2017).
37. E. Takou, A. C. Tasolamprou, O. Tsilipakos, *et al.*, "Dynamic anapole in metasurfaces made of sculptured cylinders," Phys. Rev. B **100**(8), 085431 (2019).
38. A. G. Lamprianidis and A. E. Miroshnichenko, "Excitation of nonradiating magnetic anapole states with azimuthally polarized vector beams," Beilstein J. Nanotechnol. **9**, 1478–1490 (2018).
39. A. Basharin, V. Chuguevsky, N. Volsky, *et al.*, "Extremely high q-factor metamaterials due to anapole excitation," Phys. Rev. B **95**(3), 035104 (2016).
40. J. Hu, W. Bai, C. Tan, *et al.*, "Highly electric field enhancement induced by anapole modes coupling in the hybrid dielectric-metal nanoantenna," Opt. Commun. **511**, 127987 (2022).
41. P. Kapitanova, E. Zanganeh, N. Pavlov, *et al.*, "Seeing the unseen: Experimental observation of magnetic anapole state inside a high-index dielectric particle," Ann. Phys. **532**(12), 2000293 (2020).
42. E. Takou, A. Tasolamprou, O. Tsilipakos, *et al.*, "Anapole tolerance to dissipation losses in thermally tunable water-based metasurfaces," Phys. Rev. Appl. **15**(1), 014043 (2021).
43. S. K. Das, C. Schwanke, A. Pfuch, *et al.*, "Highly efficient thg in tio2 nanolayers for third-order pulse characterization," Opt. Express **19**(18), 16985–16995 (2011).
44. C. Taudt, *Polymer Characterization* (Springer Fachmedien Wiesbaden, Wiesbaden, 2022), pp. 89–121. See Chapter 4, Eq. (4.38).
45. R. Ruppin, "Electromagnetic energy density in a dispersive and absorptive material," Phys. Lett. A **299**(2-3), 309–312 (2002).
46. P.-G. Luan, "Power loss and electromagnetic energy density in a dispersive metamaterial medium," Phys. Rev. E **80**(4), 046601 (2009).
47. F. Nunes, T. Vasconcelos, M. Bezerra, *et al.*, "Electromagnetic energy density in dispersive and dissipative media," J. Opt. Soc. Am. B **28**(6), 1544–1552 (2011).
48. A. F. Oskooi, D. Roundy, M. Ibanescu, *et al.*, "Meep: A flexible free-software package for electromagnetic simulations by the fdtd method," Comput. Phys. Commun. **181**(3), 687–702 (2010).
49. C. Jin, B. Liu, Z. Lei, *et al.*, "Structure and photoluminescence of the tio2 films grown by atomic layer deposition using tetrakis-dimethylamino titanium and ozone," Nanoscale Res. Lett. **10**(1), 95 (2015).
50. J. L. Vazquez-Arce, T. Suta, B. Fodor, *et al.*, "Optical properties of tio2 grown by atomic layer deposition using various oxidizing agents: The ellipsometry analysis of absorption properties," Adv. Mater. Interfaces **11**(26), 2400269 (2024).
51. E. Hedl, Željko Samec, V. B. Bregović, *et al.*, "A coating design to rule them all: asymmetric fabry–perot resonators with diverse optical response based on au films with high-temperature sensitivity," Opt. Lett. **49**(15), 4102–4105 (2024).
52. H. Amick, M. Gendreau, T. Busch, *et al.*, "Evolving criteria for research facilities: vibration," Proc. SPIE **5933**, 16–28 (2005).
53. A. Miranda, M. Martins, and P. A. A. De Beule, "Simultaneous differential spinning disk fluorescence optical sectioning microscopy and nanomechanical mapping atomic force microscopy," Rev. Sci. Instrum. **86**(9), 093705 (2015).
54. R. W. Boyd, "Chapter 2 - wave-equation description of nonlinear optical interactions," in *Nonlinear Optics (Third Edition)*, R. W. Boyd, ed. (Academic Press, Burlington, 2008), pp. 69–133, third edition ed.
55. S. O. Kasap, *Optoelectronics and Photonics: Principles and Practices* (Pearson Education, 2013), 2nd ed. Ch. 1: p. 31; p. 33 (Fig. 18); p. 34 for SiO2.
56. V. Atanassova, G. Yankov, K. Shumanov, *et al.*, "Sub-ablative femtosecond laser modification of the nonlinear optical response of amorphous tio2 thin films," Coatings **16**(2), 220 (2026).
57. R. Vilarinho, H. Mahmoodnia, C. Maibohm, *et al.*, "Volatile organic compound photoionization utilizing tunable quasi-monochromatic vacuum ultraviolet down to 120 nm obtained by third harmonic generation from a suspended dielectric nanomembrane," Opt. Express **33**(12), 26342–26355 (2025).
58. V. Garcia de Los Rios, M. Hernandez, J. Arano-Martinez, *et al.*, "Tunable optical kerr transmittance and structured light by a nanostructured tio2 bilayer," J. Opt. Soc. Am. B **42**(2), 292–301 (2025).